The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 February 2024.

INCH-POUND

MIL-PRF-19500/599E
w/AMENDMENT 4
27 November 2023
SUPERSEDING
MIL-PRF-19500/599E
w/AMENDMENT 3
19 October 2018

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, QUAD, FIELD EFFECT,
P-CHANNEL, SILICON, 14-PIN DUAL INLINE PACKAGE, TYPE 2N7335,
QUALITY LEVELS JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for quad P-channel, enhancement-mode, MOSFET, power transistor with avalanche energy ratings (E_{AS} and E_{AR}) and maximum avalanche current (I_{AR}). Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device type as specified in MIL-PRF-19500. Two levels of product assurance (JANHC and JANKC) for each unencapsulated device type die (element evaluation) as specified in MIL-PRF-19500.

1.2 Physical dimensions. The device package styles are as follows: 14 pin dual-in line package in accordance with figure 1 and unencapsulated die in accordance with figure 2 for device types JANHC and JANKC.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

P_T (1) T_C = +25°C (free air)		V_{GS}	I_{D1} (2) (3)	I_{D2} (2)	I_S	E_{AS}	E_{AR}
(1 die)	(4 die)		T_C = +25°C	T_C = +100°C			
W	W	V dc	A dc	A dc	A dc	mJ	mJ
1.4	2.5	±20	-.75	-.50	-.75	75	.14

I_{AR} (2)	I_{DM} (4)	T_J and T_{STG}	Max $r_{DS(on)}$ (1) V_{GS} = 10 V dc, I_D = I_{D2}		$R_{\theta JA1}$ maximum (1 die)	$R_{\theta JA2}$ maximum (4 die)	$R_{\theta JC}$ maximum (1 die)
			T_J = +25°C	T_J = +150°C			
A	A(pk)	°C	Ω	Ω	°C/W	°C/W	°C/W
-.75	-3.0	-55 to +150	1.4	2.5	90	50	17

See footnotes on next page.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 5961

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1.3 Maximum ratings – Continued.

(1) Derate linearly 0.011 W/°C for T_C > +25°C for 1 die.
(2) The following formula derives the maximum theoretical I_D limit. I_D is limited by package and internal wires and may be limited by pin diameter: $$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$
(3) See figure 3, maximum drain current graph.
(4) I_{DM} = 4 x I_{D1} as calculated in note 2.

1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = 1 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA		Max I_{DSS1} V_{GS} = 0 V V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(on)1}$ (1) V_{GS} = 10 V dc $I_D = I_{D2}$ T_J = +25°C
	V dc	V dc		µA dc	Ω
		Min	Max		
2N7335	-100	-2.0	-4.0	-25	1.4

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

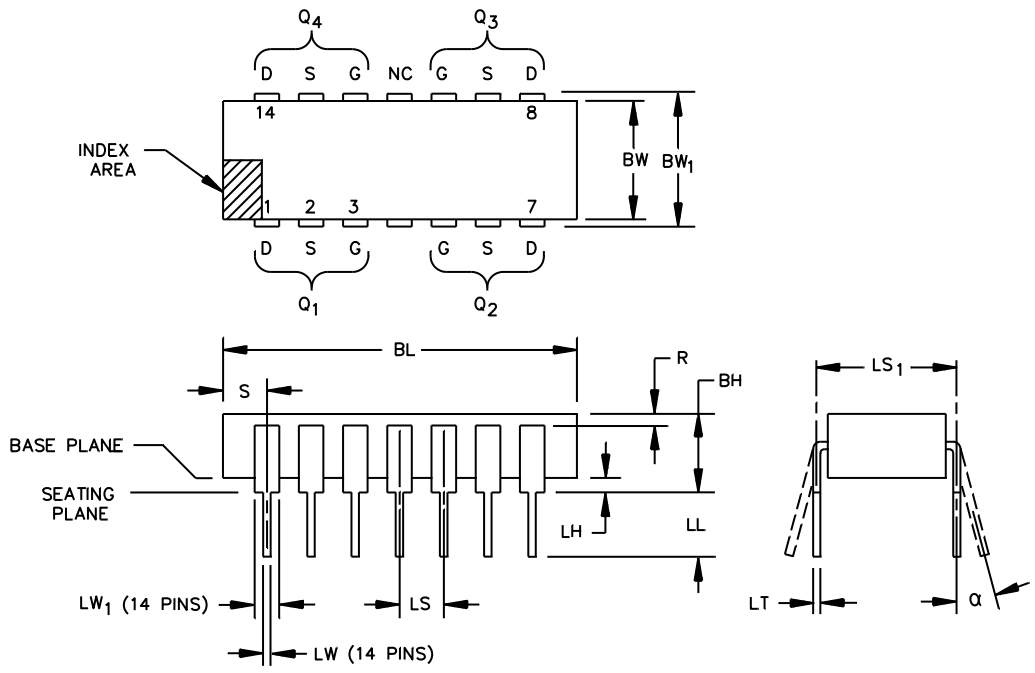
1.5.2.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet is "7335".

1.5.3 Suffix symbols. Suffix symbols are not applicable for this specification sheet.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

1.5.5 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A" (see 6.4 and figure 2).

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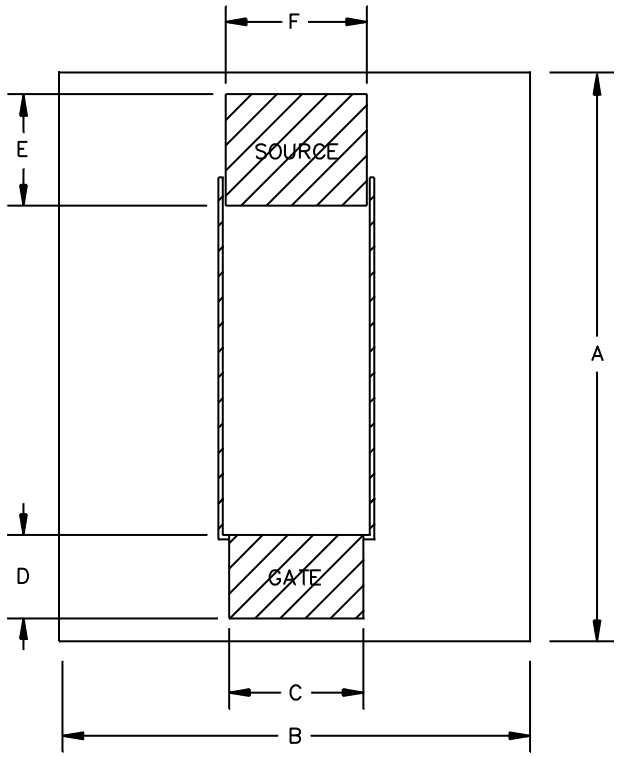
Symbol	Dimensions				Notes
	In	In	mm	mm	
	Min	Max	Min	Max	
BH	.105	.175	2.67	4.45	4
BL	.690	.770	17.53	19.56	5
BW	.280	.310	7.11	7.87	5
BW_1	.290	.325	7.37	8.26	6
LH	.025	.055	0.64	1.40	4, 7
LL	.125	.175	3.18	4.45	4
LT	.008	.012	0.203	0.305	

Symbol	Dimensions				Notes
	In	In	mm	mm	
	Min	Max	Min	Max	
LS	.100 TP		2.54 TP		8
LS_1	.300 TP		7.62 TP		
LW	.015	.021	0.381	0.533	7
LW_1	.038	.060	0.97	1.52	
R	.010		0.25		
S	.030	.095	0.76	2.41	
α	0°	15°	0°	15°	9

NOTES:
1. Dimensions are in inches. Millimeters are for general information only.
2. Pin-out: G = gate, S = source, D = drain, and NC = not connected.
3. Index area: A notch or a pin 1 identification mark shall be located adjacent to pin 1. The manufacturer's identification shall not be used as a pin 1 identification mark.
4. This dimension shall be measured with the device seated in JEDEC Seating Plane Gauge GS–003.
5. This dimension includes allowance for glass overrun, meniscus, and lid to base mismatch.
6. Lead center when α is 0°. Dimension BW_1 shall be measured at the centerline of the leads.
7. Outlines on which the seating plane is coincident with the base plane (dimension LH = 0), terminal standoffs are not required, and dimension LW_1 may equal dimension LW along any part of the lead above the seating/base plane.
8. Terminals within .005 inch (0.13 mm) radius of True Position (TP) at seating gauge plane with maximum material condition and unit installed.
9. Dimension α applies to spread leads prior to installation.

FIGURE 1. Physical dimensions and configuration of MS-030AB (see 6.8).

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Letter	Dimensions			
	Minimum	Maximum	Minimum	Maximum
A	.086	.096	2.18	2.44
B	.065	.075	1.65	1.91
C	.020	.030	0.51	0.76

Letter	Dimensions			
	Minimum	Maximum	Minimum	Maximum
D	.014	.024	0.36	0.61
E	.013	.023	0.33	0.58
F	.020	.030	0.51	0.76

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Unless otherwise specified, tolerance is ± .005 inch (0.13 mm).
3. The physical characteristics of the die thickness are as follows: The back metals are chromium, nickel, and silver (Cr, Ni, and Ag). The top metal is aluminum (Al).
4. The back contact of the die (not shown) is the drain.
5. Die thickness is .0187 inch (0.475 mm) ±.0005 inch (0.013 mm).

FIGURE 2. Physical dimensions of die JANHCA and JANKCA (A version).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500, and as follows:

I_{AS}	Rated avalanche current, nonrepetitive.
nC	nano Coulomb.

3.4 Interface and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (MS-030AB) and figure 2 (unencapsulated die) herein.

3.4.1 Lead finish. Unless otherwise specified, the lead finish shall be solderable as defined in MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Pin out. The pin out of the device types shall be as shown on figures 1 and 2.

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3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.5.1 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, R ≤ 100 kΩ, whenever bias voltage is to be applied drain to source.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4, tables I, II and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.2.1 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be as specified in MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification that did not request the performance of table III tests, the tests specified in table III herein shall be performed by the first inspection lot of this revision to maintain qualification.

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4.3 Screening.

4.3.1 Screening of packaged devices (quality levels JANS, JANTX and JANTXV only). Screening of packaged devices shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for Quality level JANS (1) (2)	Measurements for Quality levels JANTX and JANTXV (1) (2)
(3)	Gate stress test (see 4.3.1.1)	Gate stress test (see 4.3.1.1)
(3) (4)	Method 3470 of MIL-STD-750, (see 4.3.1.2) optional	Method 3470 of MIL-STD-750, (see 4.3.1.2) optional
3c (3)	Method 3161 of MIL-STD-750, (see 4.3.1.3)	Method 3161 of MIL-STD-750, (see 4.3.1.3)
9	See table I, subgroup 2 herein	See table I, subgroup 2 herein
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	Subgroup 2 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater	Subgroup 2 of table I herein
12	Method 1042 of MIL-STD-750, test condition A, t = 240 hours	Method 1042 of MIL-STD-750, test condition A
13	Subgroups 2 and 3 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value $\Delta V_{GS(th)1}$ = ±20 percent of initial value	Subgroup 2 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value $\Delta V_{GS(th)1}$ = ±20 percent of initial value

(1) At the end of the test program, I_{GSSF1}, I_{GSSR1} and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1} and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime before screen 9.
(4) This test method in no way implies a repetitive single pulse avalanche energy rating. This test need not be performed in group A when performed as a screen.

4.3.1.1 Gate stress test. Apply V_{GS} = +30 V minimum for t = 250 µs minimum.

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4.3.1.2 Single pulse avalanche energy (E_{AS}). The single pulse avalanche energy capability shall be performed in accordance with method 3470 of MIL-STD-750. The following details shall apply:

a.	Peak current (I_{AS})	I_{D1}.
b.	Peak gate voltage (V_{GS})	-10 V.
c.	Gate to source resistor (R_{GS})	$25 \leq R_{GS} \leq 200\Omega$.
d.	Initial case temperature	+25°C +10°C, -5°C.
e.	Inductance	$\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{V_{BR} - V_{DD}}{V_{BR}}\right]$ mH minimum.
f.	Number of pulses to be applied	1 pulse minimum.
g.	Supply voltage (V_{DD})	-25 V minimum.

4.3.1.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, (and V_H where appropriate). (See figure 4 herein for thermal response curves.) See table III, group E, subgroup 4 herein.

4.3.2 Screening of unencapsulated die (JANHC and JANKC). Screening of JANHC and JANKC unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. The burn-in duration for JANKC level shall follow the JANS requirements; the JANHC level shall follow the JANTX requirements of table E-IV of MIL-PRF-19500. As a minimum, die shall be 100 percent probed in accordance with group A.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. Alternate conformance inspection flow in accordance with appendix E of MIL-PRF-19500 is allowed.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, and table I herein. Delta measurements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and herein. Delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Quality level JANS (table E-VIA of MIL-PRF-19500).

	Subgroup	Method	Conditions
	B3	1051	Condition G.
*	B5	1042	Condition B (Steady-state gate bias); V_{GS} = 80 percent of rated, T_A = +175°C, t = 24 hours.
*	B5	1042	Condition A (Steady-state reverse bias); V_{DS} = 80 percent of rated, T_A = +175°C, t = 120 hours. Read and record $V_{BR(DSS)}$ (pre and post) at I_D = 1 mA.
	B6	3161	See 4.5.2.

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4.4.2.2 Quality levels JAN, JANTX, and JANTXV (table E-VIB of MIL-PRF-19500).

	Subgroup	Method	Conditions
	B2	1051	Condition G.
*	B3	1042	Condition A (Steady-state reverse bias), V_{DS} = 80 percent of rated, T_A = +150°C, t = 160 hours. Condition B (Steady-state gate bias), V_{GS} = 80 percent of rated, T_A = +150°C, t = 24 hours.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows herein. Delta requirements shall be in accordance with the applicable steps of table II herein.

	Subgroup	Method	Conditions
	C2	2036	Condition E. The sampling plan applies to the number of leads tested. A minimum of three devices shall be tested.
	C5	3161	See 4.5.2.
*	C6	1042	Condition A (Steady-state reverse bias), V_{DS} = 80 percent of rated, T_A = +150°C, t = 340 hours. Electrical measurements in accordance with table I, subgroup 2 herein. Condition B (Steady-state gate bias), V_{GS} = 80 percent of rated, T_A = +150°C, t = 24 hours.

4.4.4 Group E inspection (except JANHC and JANKC). Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Delta requirements shall be in accordance with the applicable steps of table II herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse response measurements. The conditions for pulse response measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JA1}$ maximum = 90°C/W, for each die. $R_{\theta JA1}$ shall be performed on each die. The following details shall apply:

a.	I_M measuring current	10 mA.
b.	I_H drain heating current	0.15 A minimum.
c.	t_H heating time	Steady-state (see method 3161 of MIL-STD-750 for definition).
d.	V_H drain-source heating voltage	15 V minimum.
e.	t_{MD} measurement time delay	30 to 60 µs.
f.	t_{SW} sample window time	10 µs maximum.

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TABLE I. Group A inspection.

	Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
		Method	Conditions		Min	Max	
	Subgroup 1						
	Visual and mechanical inspection	2071					
	Subgroup 2						
	Thermal impedance 2/	3161	See 4.3.1.3	$Z_{\theta JC}$		10	°C/W
	Breakdown voltage, drain-to-source	3407	Bias condition C, V_{GS} = 0V, I_D = -1 mA dc	$V_{(BR)DSS}$	-100		V dc
	Gate-to-source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = -0.25 mA	$V_{GS(th)1}$	-2.0	-4.0	V dc
*	Gate current (forward)	3411	Bias condition C, V_{GS} = +20V dc, V_{DS} = 0 V dc	I_{GSSF1}		+100	nA dc
*	Gate current (reverse)	3411	Bias condition C, V_{GS} = -20 V dc, V_{DS} = 0 V dc	I_{GSSR1}		-100	nA dc
	Drain current	3413	Bias condition C, V_{GS} = 0 V dc, V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		-25	µA dc
	Static drain to source on-state resistance	3421	Bias condition A, V_{GS} = -10 V dc, I_D = rated I_{D2} (see 1.3), pulsed (see 4.5.1)	$r_{DS(on)1}$		1.4	Ω
	Forward voltage	4011	Condition A, V_{GS} = 0 V dc, I_D = rated I_{D1}	V_{SD}		5.5	V
	Subgroup 3						
	High temperature operation:		$T_C = T_J$ = +125°C				
	Gate current	3411	Bias condition C, V_{DS} = 0 V dc, V_{GS} = +20V dc and -20 V dc	I_{GSS2}		±200	nA dc
	Drain current	3413	Bias condition C, V_{GS} = 0 V dc, V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}		-0.25	mA dc
	Static drain-to-source on-state resistance	3421	Bias condition A, V_{GS} = -10 V dc, I_D = rated I_{D2}, pulsed (see 4.5.1)	$r_{DS(on)2}$		2.3	Ω

See footnote at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 3 – Continued						
Gate-to-source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = -0.25 mA	$V_{GS(th)2}$	-1.0		V dc
Low temperature operation:		$T_C = T_J$ = -55°C				
Gate-to-source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = -0.25 mA	$V_{GS(th)3}$		-5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D1}; V_{GS} = -10 V dc; R_G = 7.5 Ω; V_{DD} = 50 percent of $V_{(BR)DSS}$				
Turn-on delay time			$t_{d(on)}$		30	ns
Rise time			t_r		60	ns
Turn-off delay time			$t_{d(off)}$		70	ns
Fall time			t_f		80	ns
Subgroup 5						
Single pulse unclamped inductive switching 3/	3470	See 4.3.1.2, n = 116, c = 0	E_{AS}			
Electrical measurements		See subgroup 2 of this table				
Safe operating area test for power MOSFETs	3474	V_{DS} = 80 percent of rated $V_{(BR)DSS}$, I_D = 0.5 A, t_p = 10 ms, (see figure 5)				
Electrical measurements		See subgroup 2 of this table				
Subgroup 6						
Not applicable						

See footnote at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{g(on)}$		15	nC
Gate to source charge			Q_{gs}		7.0	nC
Gate to drain charge			Q_{gd}		8.0	nC
Reverse recovery time	3473	Condition A, $di/dt \leq -100$ A/μs, $V_{DD} \leq -30$ V, $I_D = I_{D1}$, (see 1.3)	t_{rr}		200	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point electrical measurements, this test is required for the following subgroups:
Table E-VIA of MIL-PRF-19500, Group B, subgroups 3 and 4 (quality level JANS);
Table E-VIB of MIL-PRF-19500, Group B, subgroups 2 and 3 (quality levels JAN, JANTX, and JANTXV);
Table E-VII of MIL-PRF-19500, Group C, subgroup 2 and 6 (all quality levels);
Table E-IX of MIL-PRF-19500, Group E, subgroups 1 (all quality levels).
3/ This test need not be performed in group A if performed in screening.

TABLE II. Groups A, B and C delta measurements. 1/

Step	Inspection	MIL-STD-750		Symbol	Limits	Unit
		Method	Conditions			
1	Breakdown voltage drain to source	3407	V_{GS} = 0 V dc, I_D = -1 mA dc test condition C	$\Delta V_{(BR)DSS}$	±10 percent	
2	Drain current	3413	Bias condition C, V_{GS} = 0 V dc, V_{DS} = 80 percent of rated V_{DS}	ΔI_{DSS1}	ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater.	

1/ The delta measurements for group B, product assurance level JANS shall be as follows: In addition to the measurements specified for subgroup 5 of table E–VIA of MIL-PRF-19500, the measurements of steps 1 and 2 shall also be taken.

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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only. 1/

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Condition G, 500 cycles.	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2.	
Subgroup 2 2/			45 devices c = 0
* Steady-state reverse bias	1042	Condition A (Steady-state reverse bias), 1,000 hours.	
Electrical measurements		See table I, subgroup 2.	
* Steady-state gate bias	1042	Condition B (Steady-state gate bias), 1,000 hours.	
Electrical measurements		See table I, subgroup 2.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			
Not applicable			
Subgroup 6			11 devices
Electrostatic discharge sensitivity	1020		
Subgroup 11			22 devices c = 0
Test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors	3476		

1/ JANHC and JANKC devices are qualified in accordance with appendix G of MIL-PRF-19500.
2/ A separate sample for each test may be selected.

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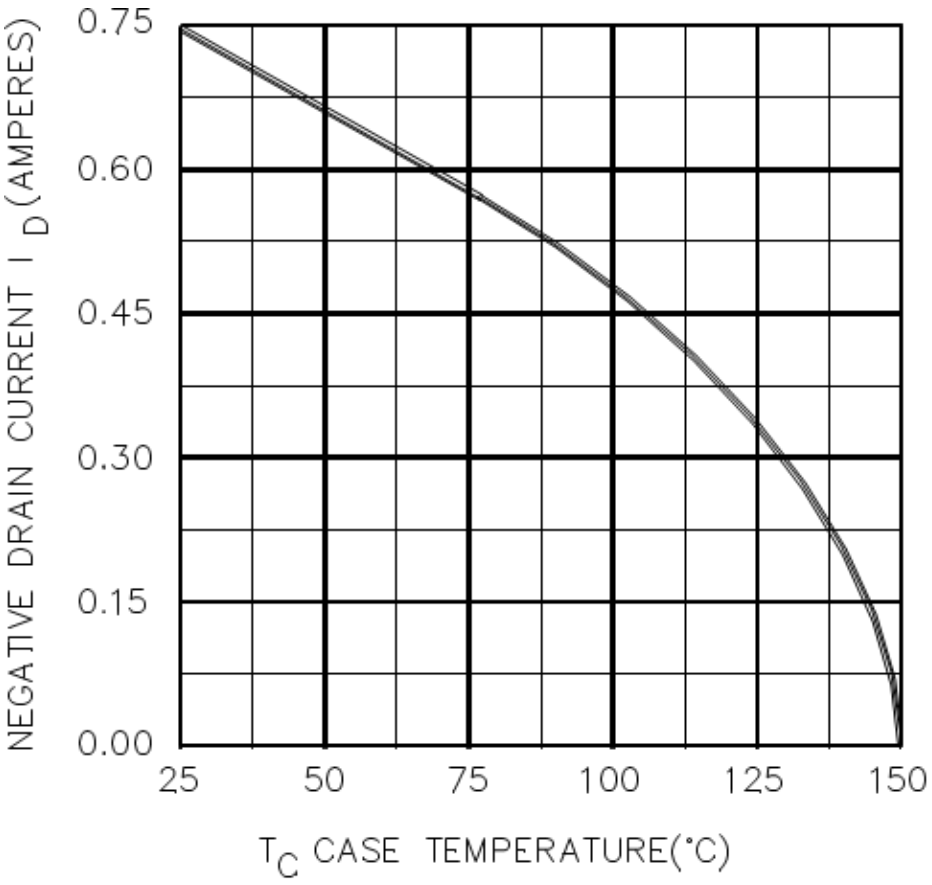
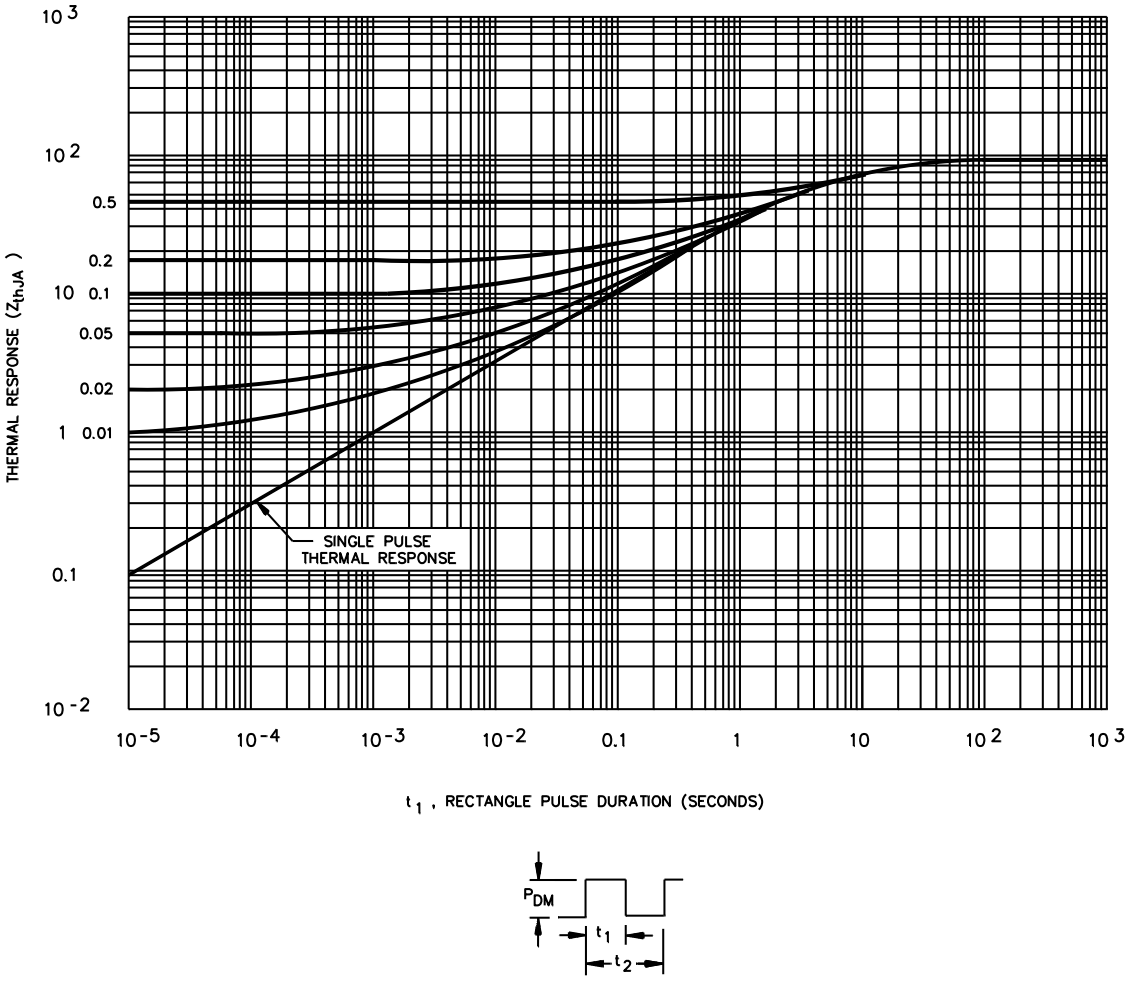


FIGURE 3. Maximum drain current vs case temperature graph.

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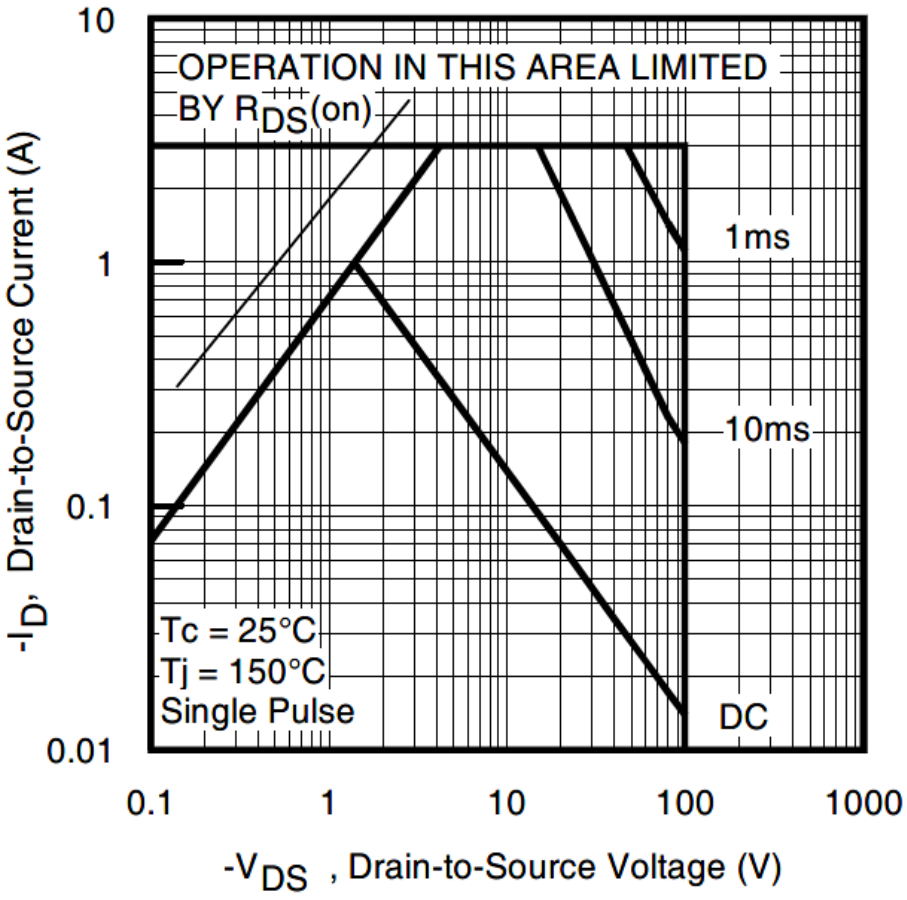


NOTES:

1. The curves are from bottom to top are: Single pulse, D = 0.01, D = 0.02, D = 0.05, D = 0.1, and D = 0.5.
2. Pulse duration magnitude = P_{DM}
3. Duty factor D = t_1 / t_2.
4. Peak $T_J = P_{DM} \times Z_{thJA} + T_C$.
5. Thermal impedance, junction to ambient = $Z_{thJA} = Z_{\theta JA}$.

FIGURE 4. Normalized transient thermal impedance.

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* FIGURE 5. Maximum safe operating area.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use . Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

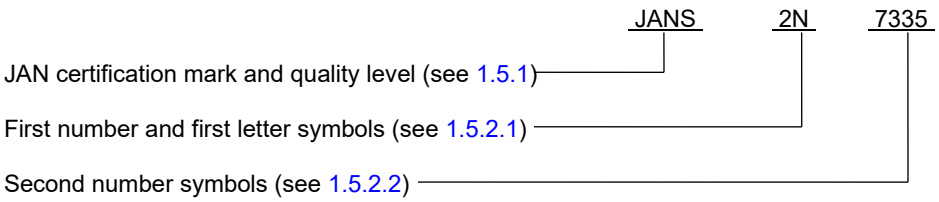
d. The complete PIN, see 1.5, 6.4 and 6.5.

e. For die acquisition, the JANHC or JANKC letter version shall be specified (see figure 2).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers' List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43216-5000 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

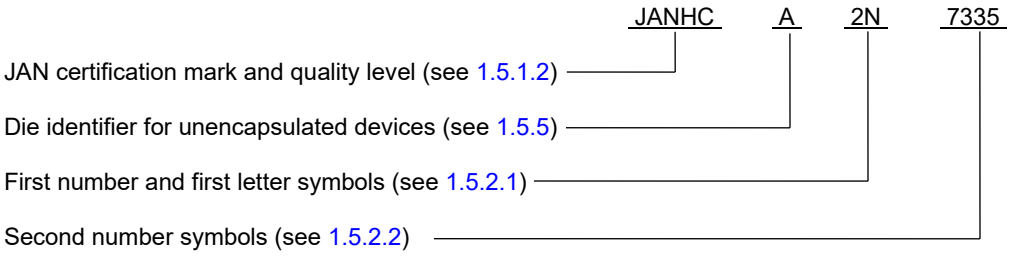
6.4 PIN construction example.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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6.4.2 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs. The following is a list of possible PINs available for encapsulated devices covered by this specification sheet.

JAN2N7335
JANTX2N7335
JANTXV2N7335
JANS2N7335

6.6 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N7335) will be identified on the QML.

PIN	Manufacturer
	69210
2N7335	JANHCA2N7335 (1) JANKCA2N7335 (1)

(1) The JANHC2N7335 and the JANKC2N7335 die were never qualified for listing on QML-19500.

6.7 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's PIN. This information in no way implies that manufacturer's PIN's are suitable for the military PIN.

Military PIN	Manufacturers' CAGE code	Manufacturers' and users' PIN
2N7335	69210	IRFG9110

6.8 Package identifier. Package designator MO-036AB used in previous issues of this specification sheet was rescinded in JEDEC Publication JEP95 "Registered and Standard Outlines for Solid state and Related Products" dated March 1997. The superseding package is identified as MS-030AB which has identical dimensions and notes.

6.9 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at "Semiconductor@dla.mil" or by facsimile (614) 693-1642 or DSN 850-6939.

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6.10 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army – CR
* Navy – SH
Air Force – 85
NASA – NA
DLA – CC

Review activities:
Army – SM
Navy – AS, MC
Air Force – 19

Preparing activity:
DLA – CC

(Project 5961-2023-105)

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